9. Why is precision navigation difficult in GPS-denied environments?
10. Describe how technologies like LiDAR and SLAM improve drone navigation accuracy.
11. What are the key design considerations in developing reliable obstacle detection and avoidance systems?
12. Explain the importance of AI and machine learning in real-time drone decision-making.
13. How does edge computing contribute to autonomous drone operations with minimal human intervention?
14. What are the key challenges in navigating global UAV regulatory frameworks?
15. Compare how different aviation authorities (FAA, EASA, and DGCA) regulate drone operations.
16. How does airspace integration differ between civilian and military drone applications?
17. Discuss the challenges of managing drone air traffic in densely populated urban areas.
18. What is the role of unmanned traffic management (UTM) systems in ensuring safe drone operation?
19. Identify the main privacy concerns associated with widespread drone surveillance.
20. How can governments balance national security with individual privacy rights in drone use?
21. Explain the challenges in assigning legal responsibility for autonomous drone operations.
22. What is the concept of the “accountability gap” in autonomous UAV systems?